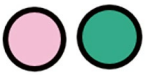
# CHAPTER 11

# Architecture Domains



This chapter is part of the PINK and GREEN reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

*This chapter builds on the architecture roles mentioned in the previous chapter. Among other things, it identifies the Segment Architect. This role is responsible for (one of) the business, data, application, and technology domains, depending on the perspective from which the organization is viewed.*

*Upon reviewing the Architecture Capability described in Chapter 9, it is evident that certain architecture roles are required within an organization depending on the capability's positioning. The TOGAF Standard distinguishes three core architecture roles that relate to the Architecture Capability.*

*There is a role for the Chief/Enterprise Architect when the Architecture Capability is used to support strategy. The Segment Architect role is focused on supporting portfolios – and in some cases (large) projects – and the Solution Architect role is primarily focused on providing support when the Architecture Capability is used to support solutions. Regardless of the architecture role practiced, an architect will have to deal with one or more architecture domains. This chapter explains them.*

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E. Jager, *Mastering the TOGAF® Standard*, https://doi.org/10.1007/979-8-8688-1814-1_11

# 11.1. Four Architecture Domains

The previous two chapters explained the use of the Architecture Capability and described the four core architecture roles as recognized by the TOGAF Standard. The relationship between the Architecture Capability, the required architecture roles, and the architecture domains has become progressively clearer.

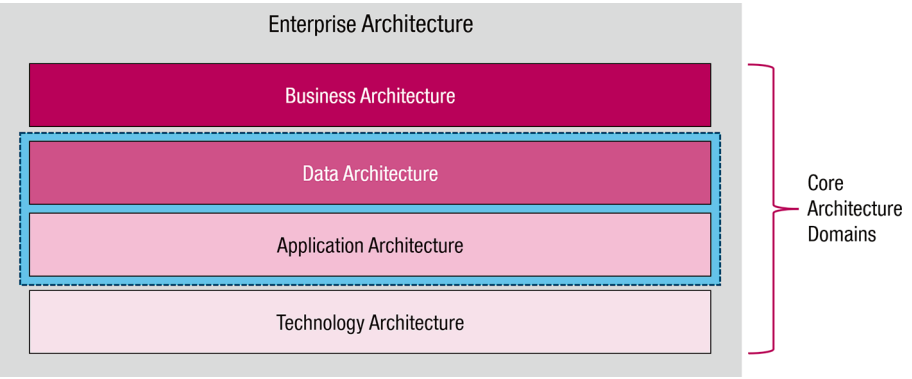



***Figure 11-1.*** *The four core architecture domains*

Figure 11-1 illustrates the four core architecture domains identified in the TOGAF Standard:

- **Business Architecture:** Defines the business strategy, governance, organization, and key business capabilities. By articulating these elements, organizations can ensure that their IT systems support and enhance business objectives.
- **Data Architecture:** Describes the structure of an organization's logical and physical data assets and data management resources. A well-defined Data Architecture facilitates effective data governance and utilization, enabling informed decision-making.

- **Application Architecture:** Provides a blueprint for individual systems to be deployed, their interactions, and their relationships to core business processes. This ensures that applications are aligned with business needs and can adapt to changing requirements.

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Data Architecture and Application Architecture are represented in the TOGAF Standard under the collective name of *Information Systems Architectures*. This is visualized in Figure 11-1 by the blue outlined area.

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- **Technology Architecture:** Details the hardware, software, and network infrastructure required to support mission-critical applications. A robust Technology Architecture underpins the reliable and efficient operation of IT services.

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The classification into four architecture domains has existed since the eighth edition of the TOGAF Standard. Due to the lack or unavailability of information on the very first versions of the TOGAF Standard, it is not possible to verify whether a different composition of architecture domains existed in the past.

What is known is that TOGAF 7 recognized only the Technology Architecture. The Business, Data, and Application Architecture domains were not yet part of this version of the framework.

It is also interesting to note that with the release of TOGAF 8 in 2002, the classification of architecture domains was as follows:

- Business Architecture
- Information Systems Architectures
  - Data or Information Architecture
  - Application Architecture
- Technology Architecture

Note in particular the addition of "or Information" in the Information Systems Architectures subheading shown in the bulleted list above. The addition of these words disappeared in later versions of the framework.

With the gradual integration between the TOGAF Standard and the BIZBOK Guide (the latter explicitly emphasizes the importance of information), a return to the usefulness and necessity of recognizing Information Architecture is gradually taking place.

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The four architecture domains mentioned above are common in architectural practice. Of course, there are methods and models that assume a slightly different division of content. For example, there is a model that divides the Business Architecture domain into two parts, *organization* and *processes*. It also assumes the existence of an Information Architecture domain instead of the Data Architecture domain. This model

is called the Interoperability Model[1] and is often used in the healthcare industry (see Figure 11-2).

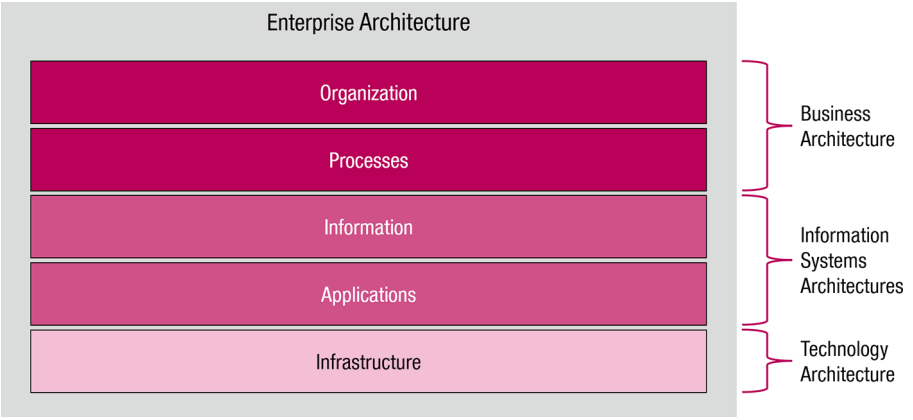



***Figure 11-2.*** *Interoperability Model (five-layer model)*

With the exception of an additional subdivision of the Business Architecture domain, which emphasizes the distinction between the organization and the processes of the organization, the model essentially assumes the four core architecture domains as described in the TOGAF Standard.

According to the TOGAF Standard, the Information Systems Architectures consists of a Data Architecture domain and an Application Architecture domain. The Data Architecture domain recognizes the *data entity* (see Chapter 5, Section 5.1.3), but does not recognize an *information concept* – an entity that is common in the BIZBOK Guide. Contrary to the TOGAF Standard, this *Business Architecture Body of Knowledge* recognizes the importance of information. From this standard, information concepts

[1] The Interoperability Model distinguishes five layers of interoperability. Each layer has its own actors, concepts, and standards. Additionally, two preconditional columns apply to all layers: legislation and regulations, and security. Source: https://nictiz.nl/wat-we-doen/zorginformatiestelsel/interoperabiliteit/lagenmodel-3/

are introduced. Information concepts are described in much more detail in the BIZBOK Guide than data entities are in the TOGAF Standard. Data entities have less granularity than information concepts. Aside from the difference between data and information, this makes the two concepts incomparable.

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The reason this book mentions information concepts is twofold.

First, because of the much more comprehensive description of the concept in the BIZBOK Guide and, second, because data is something different from information.

Data by itself does not have sufficient meaning. This means that raw data without context has little added value.

To give an example, the address *1 Primary Lane* is meaningless by itself. Without context, little can be done with it. Adding the context of *the President's home address* to 1 Primary Lane gives meaning to the raw data.

A second example is the row of numbers *22*, *24*, *26*, *21*, and *23*. This is also raw data, and without context it does not say much. When context is added to this data, such as indicating that these are the expected temperatures for next week, the data becomes meaningful. It becomes information.

This yields the following equation: Data + Context = Information.

The two examples show that information (data in context) leads to a valuable concept.

---

With the gradual movement of the TOGAF Standard toward more integration with other major frameworks and standards such as the BIZBOK Guide, it is quite possible that over time the Information Architecture domain will reappear in the TOGAF Standard, as it did in the 2008 eighth edition.

According to the TOGAF Standard, a complete Enterprise Architecture should address all four core architecture domains (Business, Data, Application, Technology). However, the realities of resource and time constraints often mean there is not enough time, funding, or resources to build a top-down, all-inclusive architecture description encompassing all four architecture domains.

Architecture descriptions are typically crafted with a clear purpose, driven by specific business objectives that steer architecture development. Defining the particular challenges the architecture description aims to address, and the questions it seeks to answer, is a crucial aspect of the initial phase of the Architecture Development Method.

For instance, if an architecture initiative's goal is to identify and assess technology options for achieving a specific capability, and the core business processes are not subject to change, a comprehensive Business Architecture might not be necessary. However, since Data, Application, and Technology Architectures are built upon the Business Architecture, it's essential to consider and comprehend the Business Architecture.

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A complete Enterprise Architecture should address all four architecture domains (Business, Data, Application, Technology) [20].

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While certain situations may necessitate developing an architecture description that doesn't encompass all four architecture domains, it is important to recognize that such an architecture cannot, by definition, be a complete Enterprise Architecture. One risk associated with this approach is a lack of consistency, leading to integration challenges. Integration

may need to occur later – introducing additional costs and risks – or the architect must clearly articulate the risks and trade-offs of not developing a complete and integrated architecture, ensuring that the organization's management understands them.

## 11.2. Difference in Domains and Layers

Over the years, the field of architecture has matured. The evolution and application of architecture frameworks has greatly contributed to this maturation. Today, more and more organizations recognize the value of architecture. Many of these organizations employ architects to structure, unify, and standardize the deployment of information systems, the execution of business processes, and the implementation of strategies.

Architects play a critical role in establishing, applying, and evolving an organization's architecture. However, organizations are not exempt from making significant contributions. In fact, they are responsible for many areas. For instance, they must define their strategy and collaborate with architects to create feasible implementation plans. Enterprise Architecture can help organizations shape their drivers, goals, objectives, and initiatives.

An architect can do more than translate organizational strategy into execution. There are several topics that cover a wide range of areas. These topics may include organizational design, business process design and optimization, and visualization of information flows within the organization. To perform their job satisfactorily, an architect must gather information on these aspects.

Enterprise Architecture, which was considered a technology-focused field around 1990, has evolved into a mature profession over the past decades. It now covers all aspects of an organization. These aspects are referred to as *architectures* or *architecture domains*. For years, people have confused architecture domains and layers, but the two are distinct.

*Architecture layers* refer to the *different levels of abstraction* within a system or Solution Architecture, typically describing how different components interact. Architecture layers thus refer to the logical divisions of a software system or application based on the *functionality* they provide. Each layer is responsible for a specific aspect of the system and communicates with adjacent layers through predefined interfaces.

Common architecture layers (see Figure 11-3) in a typical software architecture include the *presentation layer* (which manages user interaction), the *business logic layer* (which processes business rules), the *data access layer* (which interacts with databases), and the *infrastructure layer* (which supports hosting and networking). Separating the system into layers promotes modularity, maintainability, and reusability.

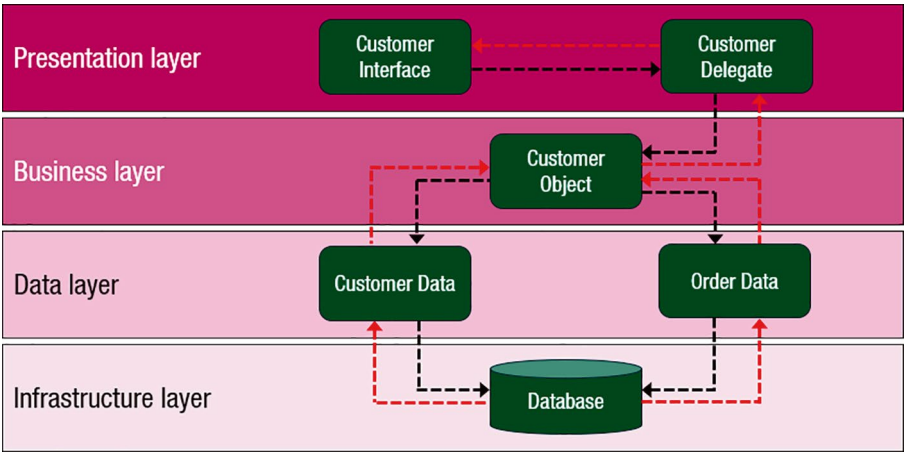



***Figure 11-3.*** *Architecture layers*

*Architecture domains*, on the other hand, are broader divisions that categorize different aspects of the overall system architecture. They represent the *areas of concern or expertise* that architects need to address while developing architecture.

The most common architecture domains are the ones that were mentioned earlier in this chapter: Business Architecture, Data Architecture, Application Architecture, and Technology Architecture. However, Security Architecture, for example, is also considered to be an architecture domain, although it is mostly called a *cross-cutting concern*. The reason for this is that Security Architecture cuts across all other architecture domains in an organization.

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Chapter 15 discusses architecture maturity models. The TOGAF Standard mentions two well-known frameworks that provide maturity models: Architecture Capability Maturity Model (ACMM) and Capability Maturity Model Integration (CMMI).

The ACMM framework lists IT security as one of the nine elements of the maturity model. However, it focuses only on the IT part of security and ignores the fact that security is a cross-cutting concern.

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Two other cross-cutting concerns are motivation and governance. All three of them are illustrated in Figure 11-4.

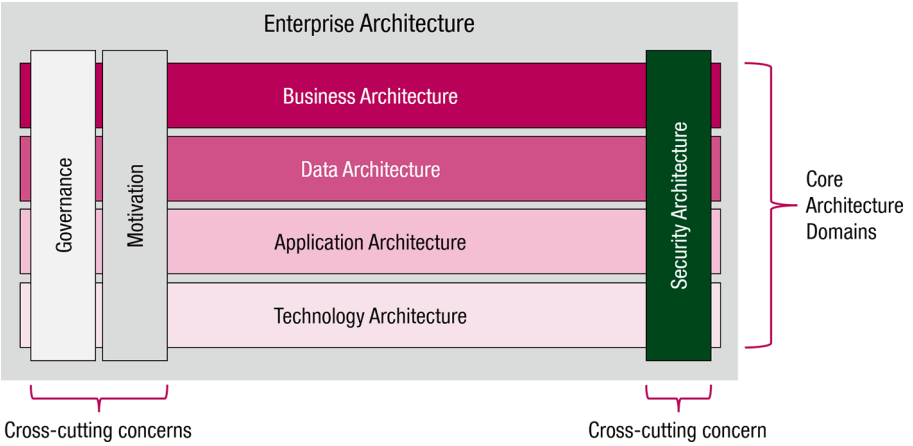



***Figure 11-4.*** *Architecture domains and cross-cutting concerns*

Each architecture domain focuses on specific concerns and constraints related to its area and contributes to the overall design of the architecture. Architecture domains categorize different concerns and perspectives that must be considered during the architectural design process. Architecture layers, on the other hand, focus on organizing the components and functionality of one or more systems within the context of an architecture domain. Architecture domains and layers are complementary concepts used to create well-structured and comprehensive architectures.

## 11.3. Mapping Roles to Domains

The four core architecture domains are directly related to the architecture roles and specializations, which in turn can be related to the specific use of the Architecture Capability. Figure 11-5 shows the relationship between the architecture roles and specializations and the corresponding architecture domains.

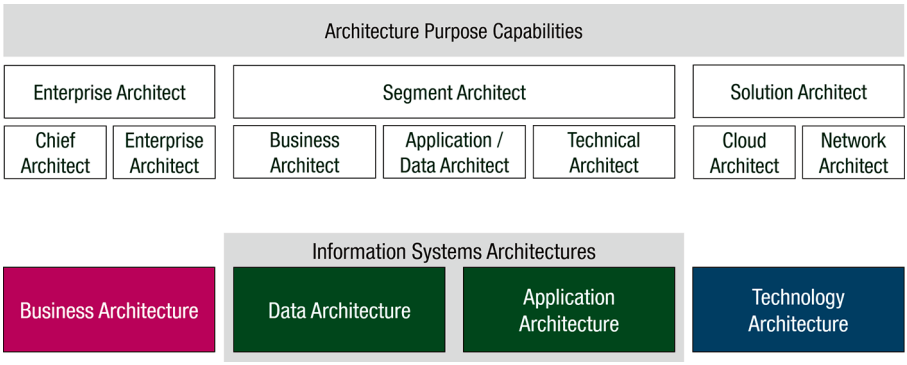



***Figure 11-5.*** *Architecture roles and specializations and the corresponding architecture domains*

A standard Enterprise Architecture consists of four architecture domains. The Enterprise Architect role is responsible for the entire ecosystem and therefore has responsibility for all four architecture domains. This is shown in Figure 11-6.

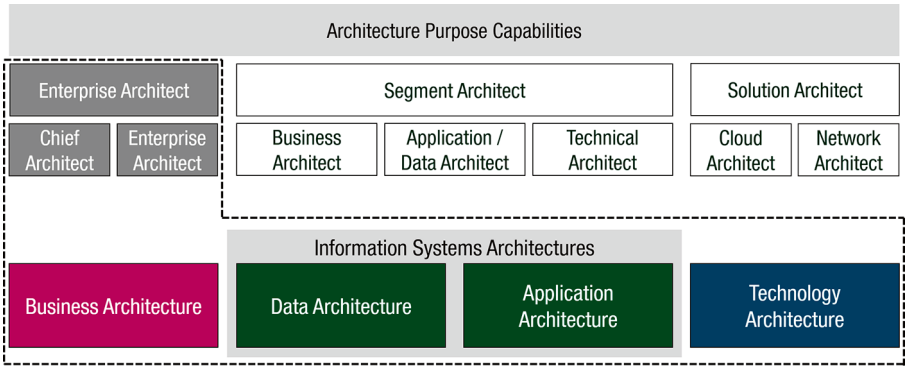



***Figure 11-6.*** *Scope of the Chief/Enterprise Architect*

When an organization has a Chief Architect, they are primarily concerned with strategic issues. If an organization does not have both a Chief Architect and an Enterprise Architect, then the work of both roles is usually performed by one and the same role, that of an Enterprise Architect.

If the focus is shifted to the role of the Segment Architect, it can be seen that this role splits into four specializations: Business, Data, Application, and Technology Architect. Each of these specializations is responsible, within the context of the architecture, for its own segment or architecture domain, Business Architecture, Data Architecture, Application Architecture, and Technology Architecture, respectively. Figure 11-7 illustrates this relationship.

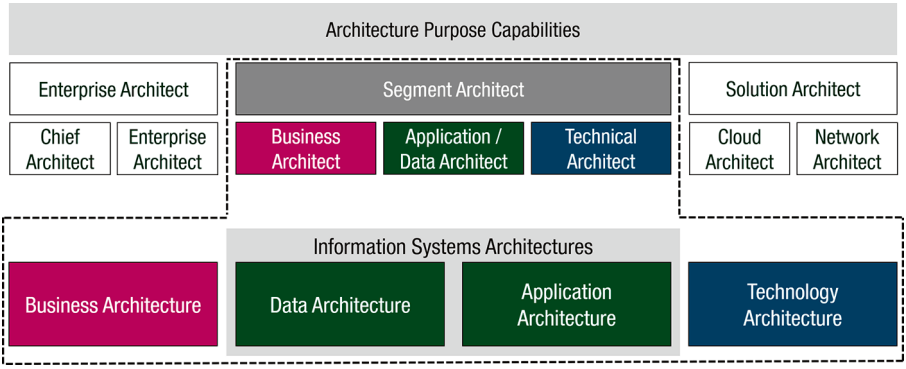



***Figure 11-7.*** *Scope of the Segment Architect*

The Solution Architect's scope of work is typically a defined project or a significant portion of a project. This is usually a technical project. Therefore, in nine out of ten scenarios, the work of a Solution Architect will take place in the context of the Technical Architecture domain. Specializations of the Solution Architect role (such as a Cloud Architect, Network Architect, or Software Architect) are all in the Technical Architecture domain (see Figure 11-8).

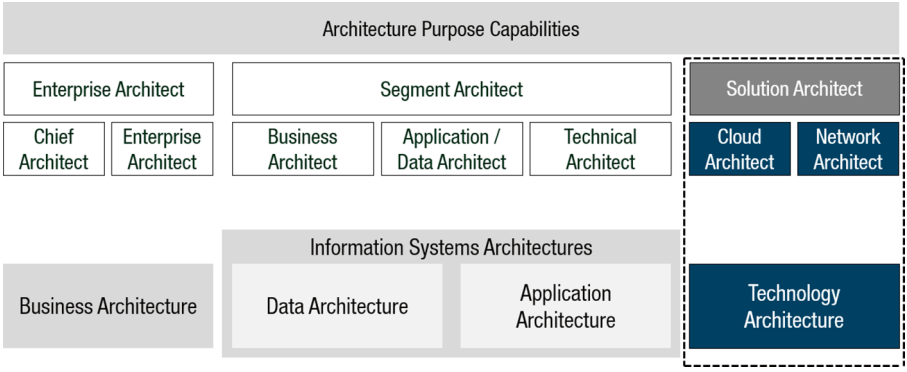



***Figure 11-8.*** *Scope of the Solution Architect*

Of course, the situations and associated examples above may differ somewhat from how they are presented here in day-to-day practice. However, it is important to recognize the distinction between architecture roles and specializations. It is also important to acknowledge that each role has its own area of interest within the context of architecture. It is therefore unwise to have false expectations of the roles mentioned.

# 11.4. Summary

This chapter has identified and described the four core architecture roles as they appear in the TOGAF Standard:

- **Business Architecture:** Defines the business strategy, governance, organization, and key business processes
- **Data Architecture:** Describes the structure of an organization's logical and physical data assets and data management resources
- **Application Architecture:** Provides a blueprint for individual systems to be deployed, their interactions, and their relationships to core business processes

- **Technology Architecture:** Details the hardware, software, and network infrastructure required to support mission-critical applications

The four core architecture domains mentioned in this chapter are common in architectural practice. There are methodologies that assume a slightly different division of content, by dividing the Business Architecture domain into two parts, organization and processes. This method is called the Interoperability Model and is often used in the healthcare industry.

A complete Enterprise Architecture should address all four core architecture domains (Business, Data, Application, Technology). While certain situations may necessitate developing an architecture description that doesn't encompass all four architecture domains, it's important to recognize that such an architecture cannot, by definition, be a complete Enterprise Architecture.

Architecture *domains*, when compared with architecture layers, are broader divisions that categorize different aspects of the overall system architecture. They represent the areas of concern or expertise that architects need to address while developing architecture. Architecture *layers* refer to the different levels of abstraction within a system or Solution Architecture. Each layer is responsible for a specific aspect of the system and communicates with adjacent layers through predefined interfaces.

The most common architecture domains are the ones that were mentioned earlier in this chapter. However, Security Architecture, for example, is also considered to be an architecture domain, although it is mostly called a cross-cutting concern. The reason for this is that Security Architecture cuts across all other architecture domains in an organization. The same is true for motivation and governance, two other cross-cutting concerns. Each architecture domain focuses on specific concerns and constraints related to its area and contributes to the overall design of the architecture. Architecture domains categorize different concerns and perspectives that must be considered during the architectural design process.

The four core architecture domains are directly related to the architecture roles and specializations, which in turn can be related to the specific use of the Architecture Capability.

The *Enterprise Architect* role is responsible for the entire ecosystem and therefore has responsibility for all four architecture domains.

From a horizontal perspective, the role of the Segment Architect can be divided into four specializations: Business Architect, Data Architect, Application Architect, and Technology Architect. Each of these specializations is responsible for its own segment or architecture domain.

The Solution Architect's scope of work is typically a defined project or a significant portion of a project. This is usually a technical project. Therefore, in nine out of ten scenarios, the work of a *Solution Architect* will take place in the context of the Technical Architecture domain.

It is important to recognize the distinction between architecture roles and specializations. It is also important to acknowledge that each role has its own area of interest within the context of architecture.